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A New Feedback Method for Dynamic Control of Manipulators

A new approach to the dynamic control of manipulators is developed from the viewpoint of mechanics. It is first shown that a linear feedback of generalized coordinates and their derivatives are effective for motion control in the large. Next, we propose a method for task-oriented coordinate control which can be easily implemented by a micro-computer and is suited to sensor feedback control. The proposed method is applicable even when holonomic constraints are added to the system. Effectiveness of the proposed method is verified by computer simulation.

1 Introduction

When we control the global motion of general manipulators, we are confronted with their nonlinear dynamics in many degrees of freedom. In much of the literature concerned with the dynamic control of manipulators, the complexity of nonlinear dynamics is emphasized and various methods that compensate all nonlinear terms in dynamics in real time are developed in order to reduce the complexity of control systems [1-3]. However, these methods require a large amount of complicated calculation so that it is difficult to implement these methods with low level controllers such as micro-computers. In addition, the reliability of these methods may be lost when a small error in computation or a small change in system's parameters occurs, since these are not considered in the control. For most industrial robots, each joint of manipulator is independently controlled by simple linear feedback [4]. However, convergence to a target position has not been sufficiently investigated for general nonlinear mechanical systems.

In this paper, we develope a new approach to the motion control of mechanical manipulators. First, we propose a simple linear state feedback control for general mechanical systems from the view point of mechanics and show that if we regard a generalized force as a control input then a linear feedback control makes the system attain to any configuration. It is also pointed out that this method is optimal in some sense.

Next, these results are extended to the case of task-oriented coordinate control which has been evolved by several authors [2], [5]. In conventional methods the data for the position described by task-oriented coordinates are transformed to a set of values for joint angles in the computer and these data are sent to each servo-system of the corresponding joint. Since in general this transformation contains an intricate nonlinear programming problem, it is very difficult to carry out the computation in real time. "Resolved Motion Rate Control" is applicable as a linear approximation method for this transformation [5], but it requires an inversion of Jacobian matrix whose calculation is quite troublesome in general.

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In the proposed control system, the manipulator is directly controlled in the task-oriented coordinate space without the transformation and the stability of the system is easily assured. Moreover, it is shown that the proposed method is still applicable even when singularity or redundancy occurs in the relation between task-oriented coordinates and manipulator's coordinates (joint angles) and some constraints which often appear in manipulator's tasks are added to the system. These advantages of the method show its robustness for the change of environment. The proposed control law includes a computation of the Jacobian matrix which requires the longest time in this method. However, since appropriate real time algorithms for calculation of the Jacobian matrix have been developed in the literature [5], the proposed method can be easily implemented by a micro-computer. Effectiveness of the proposed method is verified by computer simulation.

2 Global Position Control in the Configuration Space

We consider a general mechanical system with n degrees of freedom. The motion of the system is described by generalized coordinates $\mathbf{x} = (x_1, \ldots, x_n)^T$. In the case of manipulators, the variable x_i represents the angle of the ith joint. The kinetic energy T of the system is written as

$$T = \frac{1}{2}\dot{\mathbf{x}}^T R(\mathbf{x})\dot{\mathbf{x}} \tag{1}$$

where the matrix $R(\mathbf{x})$ is symmetric and positive definite for all $\mathbf{x}$ [6]. The generalized momentum is defined as [6]

$$\mathbf{p} = \left(\frac{\partial T}{\partial \dot{x}_1}, \ldots, \frac{\partial T}{\partial \dot{x}_n}\right)^T = \left(\frac{\partial T}{\partial \dot{\mathbf{x}}}\right)^T = R(\mathbf{x})\dot{\mathbf{x}} \tag{2}$$

It is assumed that the potential function $V(\mathbf{x})$ of the system is twice differentiable with respect to $\mathbf{x}$ and any entry of the Hessian of $V(\mathbf{x})$, $(\partial^2 V/\partial x_i \partial x_j)$, is bounded for all $\mathbf{x}$. This assumption is realized for general manipulators. The Hamiltonian H is expressed as $\mathrm{H} = T + V(\mathbf{x})$ and the equation of motion is written as [6]

$$\dot{\mathbf{x}} = \left(\frac{\partial \mathrm{H}}{\partial \mathbf{p}}\right)^T = R^{-1}(\mathbf{x})\mathbf{p} \tag{3}$$

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$$\dot{\mathbf{p}} = -\left(\frac{\partial \mathrm{H}}{\partial \mathbf{x}}\right)^T + B\mathbf{u} = -\left(\frac{\partial T}{\partial \mathbf{x}}\right)^T - \left(\frac{\partial V}{\partial \mathbf{x}}\right)^T + B\mathbf{u} \quad (4)$$

where $\mathbf{u} = (u_1, \ldots, u_n)^T$ is a vector of generalized forces corresponding to $\mathbf{x}$ and B is a nonsingular $n \times n$ matrix. Without loss of generality we assume $B = I$ (identity matrix). In this section, we consider a feedback stabilization which steers an arbitrary point in the configuration space, the space of variable $\mathbf{x}$, to the target point $\bar{\mathbf{x}} = (\bar{x}_1, \ldots, \bar{x}_n)^T$. From the view point of mechanics, position $\bar{\mathbf{x}}$ is asymptotically stable, if the potential function of the system has a minimum at $\mathbf{x} = \bar{\mathbf{x}}$ and the system is completely damped in the sense that it has a positive definite dissipation function [6]. Since we observe that the potential function of the system has great effect on both dynamic and static mechanical properties, it is natural to attempt to improve the characteristic of the system by modifying the potential function. Let $V^0(\mathbf{x})$ be a desirable potential function which is chosen in accordance with the aim of control. Let us first observe that if the control vector is set as

$$\mathbf{u} = -\left(\frac{\partial V^0}{\partial \mathbf{x}}\right)^T + \left(\frac{\partial V}{\partial \mathbf{x}}\right)^T \quad (5)$$

and it is substituted into equation (4), the potential function of the system turns out to be a desired potential function $V^0(\mathbf{x})$. Therefore, the Hamiltonian $\bar{\mathrm{H}}$ of the controlled system is described by

$$\bar{\mathrm{H}} = T + V^0(\mathbf{x}) \quad (6)$$

and the system equations whose potential function is modified are obtained by substituting a control vector

$$\mathbf{u} = -\left(\frac{\partial V^0}{\partial \mathbf{x}}\right)^T + \left(\frac{\partial V}{\partial \mathbf{x}}\right)^T + \mathbf{w} \quad (7)$$

into equations (3) and (4), as follows;

$$\dot{\mathbf{x}} = \left(\frac{\partial \bar{\mathrm{H}}}{\partial \mathbf{p}}\right)^T \quad (8)$$

$$\dot{\mathbf{p}} = -\left(\frac{\partial \bar{\mathrm{H}}}{\partial \mathbf{x}}\right)^T + \mathbf{w} \quad (9)$$

where $\mathbf{w}$ is an additional control vector.

Complete damping is realized by a linear feedback of velocity $\mathbf{w} = -Q\dot{\mathbf{x}}$. Here the matrix Q is symmetric and positive definite. If we substitute $\mathbf{w} = -Q\dot{\mathbf{x}}$ into equation (9) and differentiating $\bar{\mathrm{H}}$ along the solution trajectory of equations (8) and (9) (with $\mathbf{w} = -Q\dot{\mathbf{x}}$), we obtain

$$\dot{\bar{\mathrm{H}}} = \left(\frac{\partial \bar{\mathrm{H}}}{\partial \mathbf{x}}\right)\dot{\mathbf{x}} + \left(\frac{\partial \bar{\mathrm{H}}}{\partial \mathbf{p}}\right)\dot{\mathbf{p}} = \left(\frac{\partial \bar{\mathrm{H}}}{\partial \mathbf{x}}\right)\left(\frac{\partial \bar{\mathrm{H}}}{\partial \mathbf{p}}\right)^T - \left(\frac{\partial \bar{\mathrm{H}}}{\partial \mathbf{p}}\right)\left(\frac{\partial \bar{\mathrm{H}}}{\partial \mathbf{x}}\right)^T - \left(\frac{\partial \bar{\mathrm{H}}}{\partial \mathbf{p}}\right)Q\dot{\mathbf{x}} = -\dot{\mathbf{x}}^T Q\dot{\mathbf{x}} \quad (10)$$

This shows that the Hamiltonian $\bar{\mathrm{H}}$ plays a role of Lyapunov function if the potential function $V^0(\mathbf{x})$ is positive definite, i.e., $V^0(\mathbf{x}) > 0$ for all $\mathbf{x} \neq \bar{\mathbf{x}}$ and $V^0(\bar{\mathbf{x}}) = 0$. In addition, we assume that there is no equibrium position in configuration space except for $\bar{\mathbf{x}}$. Although $\dot{\bar{\mathrm{H}}}$ is negative semidefinite with respect to the state variables, the set $\{(\mathbf{x}^T, \mathbf{p}^T); \dot{\bar{\mathrm{H}}} = 0\}$ does not contain the entire motion of the system. Therefore the theorem of Lyapunov-La Salle-Lefshetz [7] proves the asymptotic stability of the equibrium position $(\bar{\mathbf{x}}^T, \mathbf{O}^T)$.

Furthermore, it holds that for any potential function $V^0(\mathbf{x})$ the control $\mathbf{w} = -Q\dot{\mathbf{x}}$ is optimal in the sense that the performance index

$$\mathrm{P.I.} = \int_0^\infty \frac{1}{2}\{\dot{\mathbf{x}}^T Q\dot{\mathbf{x}} + \mathbf{w}^T Q^{-1}\mathbf{w}\}dt \quad (11)$$

is minimized. This is easily shown from well-known Lyapunov-Bellman's equation of optimality [8]. In fact, let us consider again the system equations (8) and (9), and define the functions $h(\dot{x}, w)$ and $\lambda(t)$ as

$$h(\dot{\mathbf{x}}, \mathbf{w}) = \frac{1}{2}(\dot{\mathbf{x}}^T Q\dot{\mathbf{x}} + \mathbf{w}^T Q^{-1}\mathbf{w}) \quad ,$$

$$\lambda(t) = \int_t^\infty h(\dot{\mathbf{x}}(\tau), \quad \mathbf{w}(\tau))d\tau \quad (12)$$

Then Lyapunov-Bellman's equation (sufficient condition for optimality) can be written as

$$\begin{aligned}\frac{\partial \lambda}{\partial t} &= \min_{\mathbf{w}}\left\{h(\dot{\mathbf{x}}, \mathbf{w}) + \left(\frac{\partial \lambda}{\partial \mathbf{x}}, \frac{\partial \lambda}{\partial \mathbf{p}}\right)\begin{pmatrix}\dot{\mathbf{x}} \\ \dot{\mathbf{p}}\end{pmatrix}\right\} \\ &= \min_{\mathbf{w}}\left\{h(\dot{\mathbf{x}}, \mathbf{w}) + \left(\frac{\partial \lambda}{\partial \mathbf{x}}\right)\left(\frac{\partial \bar{\mathrm{H}}}{\partial \mathbf{p}}\right)^T \right. \\ &\quad \left. - \left(\frac{\partial \lambda}{\partial \mathbf{p}}\right)\left(\frac{\partial \bar{\mathrm{H}}}{\partial \mathbf{x}}\right)^T + \left(\frac{\partial \lambda}{\partial \mathbf{p}}\right)\mathbf{w}\right\} \end{aligned} \quad (13)$$

The right-hand side of this can be written as

$$\begin{aligned}&\min_{\mathbf{w}}\left\{\frac{1}{2}\mathbf{w}^T Q^{-1}\mathbf{w} + \left(\frac{\partial \lambda}{\partial \mathbf{p}}\right)\mathbf{w} + g(\lambda, \mathbf{x}, \mathbf{p})\right\} \\ &= \min_{\mathbf{w}}\left\{\frac{1}{2}\left(\mathbf{w} + Q\left(\frac{\partial \lambda}{\partial \mathbf{p}}\right)^T\right)^T Q^{-1}\left(\mathbf{w} + Q\left(\frac{\partial \lambda}{\partial \mathbf{p}}\right)^T\right) \right. \\ &\quad \left. + g'(\lambda, \mathbf{x}, \mathbf{p})\right\} \end{aligned} \quad (14)$$

Where $g(\lambda, \mathbf{x}, \mathbf{p})$ and $g'(\lambda, \mathbf{x}, \mathbf{p})$ do not contain $\mathbf{w}$ explicitly. Therefore the optimal control $\mathbf{w}^*$ should satisfy

$$\mathbf{w}^* = -Q\left(\frac{\partial \lambda}{\partial \mathbf{p}}\right)^T \quad (15)$$

To determine the optimal control $\mathbf{w}^*$, it is necessary to find the function λ which satisfies equation (13) [8]. However, it is easily seen that $\lambda = \bar{\mathrm{H}}$ satisfies equation (13). Therefore $\mathbf{w}^*$ is obtained from equation (15) as

$$\mathbf{w}^* = -Q\left(\frac{\partial \bar{\mathrm{H}}}{\partial \mathbf{p}}\right)^T = -Q\dot{\mathbf{x}} \quad (16)$$

The performance index P.I. is related to smoothness of the motion, and in view of this, the linear velocity feedback $-Q\dot{\mathbf{x}}$ is considered to be sufficiently effective as an optimal damping force.

Several types of functions are considered as a desired potential function $V^0(x)$. Among these, we propose the simplest function described by

$$V^0(\mathbf{x}) = V(\mathbf{x}) - V(\bar{\mathbf{x}}) - \left(\frac{\partial V}{\partial \mathbf{x}}\right)(\bar{\mathbf{x}})(\mathbf{x} - \bar{\mathbf{x}}) + \frac{1}{2}(\mathbf{x} - \bar{\mathbf{x}})^T W(\mathbf{x} - \bar{\mathbf{x}}) \quad (17)$$

where the matrix W is $n \times n$ symmetric and positive definite. This potential function is realized by linear position feedback plus constant bias, that is

$$\left(\frac{\partial V}{\partial \mathbf{x}}\right)^T - \left(\frac{\partial V^0}{\partial \mathbf{x}}\right)^T = \left(\frac{\partial V}{\partial \mathbf{x}}\right)^T(\bar{\mathbf{x}}) - W(\mathbf{x} - \bar{\mathbf{x}}) \quad (18)$$

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The positive definiteness of $V^0(\mathbf{x})$ is satisfied, if $V^0(\mathbf{x})$ is a convex function. To show this, we obtain, by differentiating $V^0(\mathbf{x})$ of equation (17) with respect to $\mathbf{x}$,

$$\left(\frac{\partial V^0}{\partial \mathbf{x}}\right)^T = \left(\frac{\partial V}{\partial \mathbf{x}}\right)^T (\mathbf{x}) - \left(\frac{\partial V}{\partial \mathbf{x}}\right)^T (\bar{\mathbf{x}}) + W(\mathbf{x}-\bar{\mathbf{x}}) \quad (19)$$

and

$$V^0(\bar{\mathbf{x}}) = 0, \quad \left(\frac{\partial V^0}{\partial \mathbf{x}}\right)^T (\bar{\mathbf{x}}) = \mathbf{O}.$$

Furthermore, differentiating the equation (19) yields

$$\left(\frac{\partial^2 V^0}{\partial x_i \partial x_j}\right)(\mathbf{x}) = \left(\frac{\partial^2 V}{\partial x_i \partial x_j}\right)(\mathbf{x}) + W \quad (20)$$

Since each $(\partial^2 V/\partial x_i \partial x_j)(\mathbf{x})$ is assumed to be bounded, we can choose W so that it satisfies

$$\left(\frac{\partial^2 V}{\partial x_i \partial x_j}\right)(\mathbf{x}) + W > 0 \quad (21)$$

for all $\mathbf{x}$. If we choose W in this way, $V^0(\mathbf{x})$ becomes a strictly convex function due to (19) and (21), and has a global minimum $V^0(\bar{\mathbf{x}}) = 0$ at $\mathbf{x} = \bar{\mathbf{x}}$. Therefore, the position $(\bar{\mathbf{x}}^T, \mathbf{O}^T)$ is asymptotically stabilized in the large by linear state feedback (plus constant bias) $\mathbf{u}$ described by

$$\mathbf{u} = \left(\frac{\partial V}{\partial \mathbf{x}}\right)^T (\bar{\mathbf{x}}) - W(\mathbf{x}-\bar{\mathbf{x}}) - Q\dot{\mathbf{x}} \quad (22)$$

In particular, we can choose gain matrices W and Q as diagonal. This implies that, in spite of nonlinear coupled nature of the dynamics, global position control can be performed by a simple linear feedback with fixed gains for controlling each joint independently. However, it is not easy to analyze the transient behavior of motion. Even if the system is asymptotically stable, joint angles may make more than one revolution. Moreover, we tacitly assumed that the configuration space is an n-dimensional Euclidian space, but the configuration space should be the n-dimensional torus and the result mentioned above is not valid in this case because the function $V^0(\mathbf{x})$ of equation (17) is not continuous on the whole torus. However, we can estimate a region in the configuration space, where the solution $\mathbf{x}(t)$ of the system (3) and (4) with control of equation (7) converges to the target position without rotating more than 2π radian. To show this we introduce a set of initial positions in such a manner that any solution started from this set does not leave the set A which is defined by

$$A = \{\mathbf{x} \quad ; \quad |x_i - \bar{x}_i| \leqq \pi \quad i = 1, \ldots, n\}$$

and goes to $\bar{\mathbf{x}}$ as $t \to \infty$. Suppose that the velocity $\dot{\mathbf{x}}$ is initially zero. We define a number μ as

$$\mu = \text{minimum}[V^0(\mathbf{x}) \quad ; \quad |x_i - \bar{x}_i| = \pi \quad i = 1, \ldots, n]$$

Then, it follows from the convexty of function $V^0(\mathbf{x})$ that $V^0(\mathbf{x}) < \mu$ implies $\mathbf{x} \in A$. Since the function $\bar{\mathrm{H}}$ is decreasing along the solution of the system, $V^0(\mathbf{x}(t_0)) < \mu$ implies $V^0(\mathbf{x}(t)) < \mu$ for all $t > t_0$ and therefore $\mathbf{x}(t) \in A$. Thus the convex set $\{\mathbf{x}\,;\, V^0(\mathbf{x}) < \mu\}$ is the region which we want.

3 Position Control in the Space of Task-Oriented Coordinate

In the previous section, it is implicitly assumed that the target position is given in the form of manipulator's joint angles. However, to make a manipulator work in real time and improve the position accuracy, it is more suitable to describe a target position in the form of external variables, such as the absolute coordinates of the hand's position and the hand's orientation.

In the rest of paper, we consider a position control by a direct feedback of task-oriented coordinates and attempt to treat the task-oriented coordinate control synthetically. Since we described the system in coordinate free form, stability and optimality results obtained in the previous section can be easily extended to this case. Let $\mathbf{y}$ be an m dimensional vector of task-oriented coordinates written by

$$\mathbf{y} = (y_1, \ldots, y_m)^T = \mathbf{f}(\mathrm{x}) = (f_1, \ldots, f_m)^T \quad (23)$$

At first, we assume that the dimension m is equal to n and its Jacobian matrix is nonsingular, i.e.,

$$\det\left[\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right] = \det[J(\mathbf{x})] \neq 0 \quad (24)$$

in a certain neighborhood O of $\bar{\mathbf{y}} = \mathbf{f}(\bar{\mathbf{x}})$. Then, from the implicit function theorem [9], there is an one-to-one continuous inverse function $\mathbf{f}^{-1}$ of $\mathbf{f}$ in O. Therefore we can rewrite the Hamiltonian $\mathrm{H}(\mathbf{x}, \mathbf{p})$ by $\mathbf{y}$ and corresponding generalized momentum $\mathbf{q}$ which is written by

$$\mathbf{q} = \left(\frac{\partial T}{\partial \dot{\mathbf{y}}}\right)^T = (J^T)^{-1} R J^{-1} \dot{\mathbf{y}} \quad (25)$$

where J, R, and function V are regarded as functions of $\mathbf{y}$ by using the inverse function, i.e., $\mathbf{x} = \mathbf{f}^{-1}(\mathbf{y})$. Then, we next calculate the generalized force term corresponding to $\mathbf{y}$. Let v_i be a generalized force corresponding to y_i. The virtual displacement of $\mathbf{y}$, $\delta\mathbf{y}$, produces a virtual displacement of $\mathbf{x}$, $\delta\mathbf{x}$, which is written as

$$\delta\mathbf{x} = J^{-1}(\mathbf{x})\delta\mathbf{y} \quad (26)$$

From this equation, virtual displacement $\delta\mathbf{x}^i$ for $\delta\mathrm{y} = (0, \ldots, \delta y_i, \ldots, 0)^T$ is obtained as

$$\delta\mathbf{x}^i = \mathbf{J}^i \cdot \delta y_i \quad (27)$$

where $\mathbf{J}^i$ is the ith column of matrix J^{-1}, i.e.,

$$J^{-1} = [\mathbf{J}^1; \ldots; \mathbf{J}^i; \ldots; \mathbf{J}^n]$$

Therefore, virtual work is calculated as

$$v_i \delta y_i = \mathbf{u}^T \delta\mathbf{x}^i = \mathbf{u}^T \mathbf{J}^i \delta y_i \quad (28)$$

and from this relation we obtain a vector of generalized forces for $\mathbf{y}$ as follows,

$$\begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} \mathbf{u}^T \mathbf{J}^1 \\ \vdots \\ \mathbf{u}^T \mathbf{J}^n \end{pmatrix} = \begin{pmatrix} (\mathbf{J}^1)^T \mathbf{u} \\ \vdots \\ (\mathbf{J}^n)^T \mathbf{u} \end{pmatrix} = (J^T)^{-1} \mathbf{u} \quad (29)$$

Then, the equation of motion for $\mathbf{y}$ becomes

$$\dot{\mathbf{y}} = \left(\frac{\partial T}{\partial \mathbf{q}}\right)^T \quad (30)$$

$$\dot{\mathbf{q}} = -\left(\frac{\partial T}{\partial \mathbf{y}}\right)^T - \left(\frac{\partial V}{\partial \mathbf{y}}\right)^T + (J^T)^{-1} \mathbf{u} \quad (31)$$

In equations (30) and (31), only the term of generalized forces is different from that of equations (3) and (4). If we input $\mathbf{u}$ defined by

$$\mathbf{u} = \left(\frac{\partial V}{\partial \mathbf{x}}\right)^T - J^T[W(\mathbf{y}-\bar{\mathbf{y}}) + Q\dot{\mathbf{y}}] \quad (32)$$

into equation (31), the Hamiltonian $\bar{\mathrm{H}}$ of this system becomes

$$\bar{\mathrm{H}} = T + \frac{1}{2}(\mathbf{y}-\bar{\mathbf{y}})^T W(\mathbf{y}-\bar{\mathbf{y}}) \quad (33)$$

Differentiating $\bar{\mathrm{H}}$ with respect to time, we obtain

$$\dot{\bar{\mathrm{H}}} = -\dot{\mathbf{y}}^T Q \dot{\mathbf{y}} \quad (34)$$

This relation shows that the position $(\mathbf{y}^T, \mathbf{q}^T) = (\bar{\mathbf{y}}^T, \mathbf{O}^T)$ is

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asymptotically stable by a similar argument as stated in the previous section. Also, it can be shown that the control $\mathbf{w} = -J^T Q\dot{\mathbf{y}}$ is optimal in the sense that the performance index

$$\int_0^\infty \frac{1}{2}(\dot{\mathbf{y}}^T Q\dot{\mathbf{y}} + \mathbf{u}^T J^{-1} Q^{-1} (J^T)^{-1}\mathbf{u})\,dt$$

is minimized.

It should be noted that $\mathbf{y}$ may be partly measured by some appropriate sensors and partly calculated by the value of $\mathbf{x}$ in computer.

If the Jacobian matrix $J(\mathbf{x})$ is singular at the target position or $m < n$, i.e., the system has redundancy for the task, the equation of motion for $\mathbf{y}$ such as (30) and (31), cannot be derived. In these cases, we introduce the slack variables for convenience. To treat both cases simultaniously, assume that the rank of Jacobian matrix J is $m(<n)$ and $y_1, \ldots, y_m$ are independent variables. The vector of slack variables $\mathbf{y}_s$ is written as

$$\mathbf{y}_s = (y_{m+1}, \ldots, y_n)^T = \mathbf{f}_s(\mathbf{x})$$

which satisfies a condition,

$$\det\left[\frac{\partial \mathbf{f}_e}{\partial \mathbf{x}}\right] = \det[J_e] \neq 0 \text{ in } O \tag{35}$$

where the vector $\mathbf{f}_e = (\mathbf{f}^T, \mathbf{f}_s^T)$ and matrix J_e is related with J as

$$J_e = \begin{bmatrix} J \\ \cdots \\ \dfrac{\partial \mathbf{f}_s}{\partial \mathbf{x}} \end{bmatrix} \begin{matrix} \} m \\ \\ \} n-m \end{matrix}$$

Then the equation of motion for $\mathbf{y}_e^T = (\mathbf{y}^T, \mathbf{y}_s^T)$ is written as

$$\frac{d}{dt}\begin{bmatrix} \mathbf{y} \\ \mathbf{y}_s \end{bmatrix} = \begin{bmatrix} \left(\dfrac{\partial H}{\partial \mathbf{q}}\right)^T \\ \left(\dfrac{\partial H}{\partial \mathbf{q}_s}\right)^T \end{bmatrix} \tag{36}$$

$$\frac{d}{dt}\begin{bmatrix} \mathbf{q} \\ \mathbf{q}_s \end{bmatrix} = -\begin{bmatrix} \left(\dfrac{\partial H}{\partial \mathbf{y}}\right)^T \\ \left(\dfrac{\partial H}{\partial \mathbf{y}_s}\right)^T \end{bmatrix} + (J_e^T)^{-1}\mathbf{u} \tag{37}$$

where vector $\mathbf{q}_s$ is a generalized momentum for $\mathbf{y}_s$. Now we consider an asymptotic stabilization with respect to $(\mathbf{y}^T, \mathbf{q}^T, \mathbf{q}_s^T)$. In this case, the control $\mathbf{u}$ of equation (32) cannot damp for the direction of $\mathbf{y}_s$ axis. Since we assume that $\mathbf{y}_s$ and $\dot{\mathbf{y}}_s$ are unknown vectors, the damping force such as $-J_e^T Q\dot{\mathbf{y}}_e$ cannot be realized. Therefore, we adopt $-Q\dot{\mathbf{x}}$ as a damping force instead of $-J_e^T Q\dot{\mathbf{y}}_e$ and set the control $\mathbf{u}$ as

$$\mathbf{u} = \left(\frac{\partial V}{\partial \mathbf{x}}\right)^T - J^T W(\mathbf{y} - \bar{\mathbf{y}}) - Q\dot{\mathbf{x}} \tag{38}$$

where W is an $m \times m$ positive definite matrix while Q is an $n \times n$ positive definite matrix. Since the term $-Q\dot{\mathbf{x}}$ can be written as $-(J_e^T)(J_e^T)^{-1} QJ_e^{-1}\mathbf{y}_e$, we can regard it as the damping force which is derived from the dissipation function

$$\dot{\mathbf{y}}_e^T (J_e^T)^{-1} Q (J_e^T)\dot{\mathbf{y}}_e$$

in the space of $\mathbf{y}_e$. We consider the asymptotic stability of the position by means of the linearized system of equations (36) and (37). Substituting $\mathbf{u}$ of equation (38) into equation (37) and linearlizing equations (36) and (37) at $(\mathbf{y}^T, \mathbf{y}_s^T, \mathbf{q}^T, \mathbf{q}_s^T) = (\bar{\mathbf{y}}^T, \bar{\mathbf{y}}_s^T, \mathbf{O}^T, \mathbf{O}^T)$, we obtain

$$\frac{d}{dt}\begin{bmatrix} \delta\mathbf{y} \\ \delta\mathbf{y}_s \end{bmatrix} = \begin{bmatrix} J_e R^{-1} J_e^T \end{bmatrix}\begin{bmatrix} \mathbf{q} \\ \mathbf{q}_s \end{bmatrix} = \begin{bmatrix} \tilde{R}_{11} & \tilde{R}_{12} \\ \tilde{R}_{12}^T & \tilde{R}_{22} \end{bmatrix}\begin{bmatrix} \mathbf{q} \\ \mathbf{q}_s \end{bmatrix} \tag{39}$$

$$\frac{d}{dt}\begin{bmatrix} \mathbf{q} \\ \mathbf{q}_s \end{bmatrix} = -\begin{bmatrix} W & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} \delta\mathbf{y} \\ \delta\mathbf{y}_s \end{bmatrix} - \begin{bmatrix} (J_e^T)^{-1} QJ_e^{-1} \end{bmatrix}\begin{bmatrix} \delta\dot{\mathbf{y}} \\ \delta\dot{\mathbf{y}}_s \end{bmatrix} \tag{40}$$

where $(\delta\mathbf{y}^T, \delta\mathbf{y}_s^T) = (\mathbf{y}^T - \bar{\mathbf{y}}^T, \mathbf{y}_s^T - \bar{\mathbf{y}}_s^T)$, and $\bar{\mathbf{y}}_s^T$ is an appropriate vector. It may be easily seen that the vector $\delta\mathbf{y}_s$ can be pulled out from the system of equations (39) and (40). Consequently, we obtain the reduced system of equations (39) and (40) as follows,

$$\frac{d}{dt}\begin{bmatrix} \delta\mathbf{y} \\ \mathbf{q} \\ \mathbf{q}_s \end{bmatrix} = \left[\begin{array}{c|cc} 0 & \tilde{R}_{11} & \tilde{R}_{12} \\ \hline -W & -(J_e^T)^{-1} QR^{-1} J_e^T & \\ \hline 0 & & \end{array}\right]\begin{bmatrix} \delta\mathbf{y} \\ \mathbf{q} \\ \mathbf{q}_s \end{bmatrix} \tag{41}$$

If we differentiate the function $\bar{H}$ of equation (33) with $m \times m$ matrix W along the solution trajectory of the system (46), we obtain

$$\dot{\bar{H}} = -(\mathbf{q}^T, \mathbf{q}_s^T) J_e R^{-1} QR^{-1} J_e^T \begin{pmatrix} \mathbf{q} \\ \mathbf{q}_s \end{pmatrix} = -\dot{\mathbf{y}}_e^T (J_e^{-1})^T QJ_e^{-1}\dot{\mathbf{y}}_e \tag{42}$$

Since the set $[(\delta\mathbf{y}^T, \mathbf{q}^T, \mathbf{q}_s^T); \dot{\bar{H}} = 0$; i.e., $(\mathbf{q}^T, \mathbf{q}_s^T) = \mathbf{O}^T]$ does not contain the entire trajectory except $(\delta\mathbf{y}^T, \mathbf{q}^T, \mathbf{q}_s^T) = \mathbf{O}^T$, we can conclude that the system (41) is asymptotically stable.

It should be noted that the vector $\mathbf{y}_s$ is introduced only for convenience of the above argument. The vector $\mathbf{y}_s$ converges to a certain position, with which we need not be concerned.

In this way, the control law of equation (38) is applicable for position control even when the system has redundancy for the task so far as we consider the linearized system (39) and (40). This result also shows the validity of the feedback of only part of the coordinates.

Though the asymptotic stability of the redundant system is assured only for the linearized system in this section, global asymptotic stability can be assured for the nonlinear system of equations (36) and (37), provided that the boundedness of the vector $\mathbf{y}_s$ can be assumed. It is rigorously proved by the result of literature [10].

4 Constrained Dynamics

In many of a manipulator's tasks, the dynamics of the system is attended by some constraints caused by a geometric correspondence between manipulator and objects. Opening a door, turning a crank, and some assembly tasks such as peg in hole are given as such examples [11]. Since the structure of the system is altered by these constraints, the stability of motion is not always preserved [12]. Therefore we should design a control system taking account of these constraints.

Let us consider the case that holonomic constraints are added to the system in such a way that

$$\Psi(\mathbf{x}) = (\psi_1(\mathbf{x}), \ldots, \psi_r(\mathbf{x}))^T = \mathbf{O} \tag{43}$$

It is assumed that $\psi_1(\mathbf{x}), \ldots, \psi_r(\mathbf{x})$ are independent and without loss of generality, $\mathbf{x}$ are partitioned as $\mathbf{x}^T = (\mathbf{x}_I^T, \mathbf{x}_D^T)$, $\mathbf{x}_I$ is an $n-r$ dimensional vector of the independent part and $\mathbf{x}_D$ is an r dimensional vector of the dependent part. From the equation (42), virtual displacement $\delta\mathbf{x}$ must satisfy

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$$\left(\frac{\partial \Psi}{\partial \mathbf{x}}\right)\delta\mathbf{x} = \left(\frac{\partial \Psi}{\partial \mathbf{x}_I}; \frac{\partial \Psi}{\partial \mathbf{x}_D}\right)\begin{pmatrix}\delta\mathbf{x}_I \\ \delta\mathbf{x}_D\end{pmatrix} = \mathbf{O} \tag{44}$$

From the assumption, it follows that det $[\partial\Psi/\partial\mathbf{x}_D] \neq 0$ and $\mathbf{x}_D$ is determined by $\mathbf{x}_I$, i.e., $\mathbf{x}_D = \mathbf{x}_D(\mathbf{x}_I)$. We begin with calculating the term of generalized forces. Control $\mathbf{u}$ is partitioned as $\mathbf{u}^T = (\mathbf{u}_I^T, \mathbf{u}_D^T)$ corresponding to $(\mathbf{x}_I^T, \mathbf{x}_D^T)$. From equation (44), $\delta\mathbf{x}_D$ is expressed by $\delta\mathbf{x}_I$ as

$$\delta\mathbf{x}_D = -\left(\frac{\partial \Psi}{\partial \mathbf{x}_D}\right)^{-1}\left(\frac{\partial \Psi}{\partial \mathbf{x}_I}\right)\delta\mathbf{x}_I = D\delta\mathbf{x}_I \tag{45}$$

Calculating the virtual work for a virtual displacement of the form $\delta\mathbf{x}_I = \delta\mathbf{x}^i = (0, \ldots, \delta x_i, \ldots, 0)^T$ $(i = 1, \ldots, n-r)$, we obtain the generalized force vector $\mathbf{u}_c$ as

$$\mathbf{u}_c = \mathbf{u}_I + D^T\mathbf{u}_D \tag{46}$$

Then, the equation of motion with constraint (43) is derived by rewriting the Hamiltonian by only $\mathbf{x}_I$ ($\mathbf{x}_D$ is specified by $\mathbf{x}_I$) as follows,

$$\dot{\mathbf{x}}_I = \left(\frac{\partial T_c}{\partial \mathbf{x}_I}\right)^T \tag{47}$$

$$\dot{\mathbf{p}}_I = -\left(\frac{\partial T_c}{\partial \mathbf{x}_I}\right)^T - \left(\frac{\partial V_c}{\partial \mathbf{x}_I}\right)^T + \mathbf{u}_I + D^T(\mathbf{x}_I)\mathbf{u}_D \tag{48}$$

where $\mathbf{p}_I$ is a generalized momentum for $\mathbf{x}_I$, T_c and V_c are defined as $T(\mathbf{x}, \mathbf{p}) = T_c(\mathbf{x}_I, \mathbf{p}_I)$, $V(\mathbf{x}) = V_c(\mathbf{x}_I)$. The term $(\partial V_c/\partial \mathbf{x}_I)$ is written as

$$\left(\frac{\partial V_c}{\partial \mathbf{x}_I}\right)^T = \left(\frac{\partial V}{\partial \mathbf{x}_I}\right)^T + D^T\left(\frac{\partial V}{\partial \mathbf{x}_D}\right)^T$$

Therefore, to stabilize the position $(\mathbf{x}_I^T, \mathbf{p}_I^T) = (\bar{\mathbf{x}}_I^T, \mathbf{O}^T)$ asymptotically, we set the control $\mathbf{u}$ as

$$\mathbf{u} = \begin{pmatrix}\mathbf{u}_I \\ \mathbf{u}_D\end{pmatrix} = \left(\frac{\partial V}{\partial \mathbf{x}}\right)^T(\bar{\mathbf{x}}) - \begin{pmatrix}W(\mathbf{x}_I - \bar{\mathbf{x}}_I) + Q\dot{\mathbf{x}}_I \\ \mathbf{O}\end{pmatrix} \tag{49}$$

where W and Q are $(n-r)\times(n-r)$ positive definite matrices, respectively, and W satisfies

$$\left(\frac{\partial^2 Vc}{\partial x_i \partial x_j}\right) + W > 0 \tag{50}$$

Furthermore, we can easily see that the control law of equation (22) also assures the asymptotic stability of the position $(\bar{\mathbf{x}}^T, \mathbf{O}^T)$ in constrained case, if the target position $\bar{\mathbf{x}}$ satisfies the constraint condition (43). In fact, the potential function $V_c^0(\mathbf{x}_I) = V^0(\mathbf{x}_I, \mathbf{x}_D)$ and dissipation function $\mathbf{x}_I^T\tilde{Q}\dot{\mathbf{x}}_I$ are positive definite as to $\mathbf{x}_I$ and $\dot{\mathbf{x}}_I$, respectively. Here, the matrix $\tilde{Q}$ is defined as

$$\tilde{Q} = (I, D^T)Q\begin{pmatrix}I \\ D\end{pmatrix}$$

These results obviously hold for the case of task-oriented coordinate control system, i.e., the system of equation (30) and (31) and the control law (32) or (38). Moreover, combining the argument for the redundant case stated in the previous section, we can treat practically interesting tasks. For example, when we made the manipulator turn a crank as shown in Fig. 1, we should select r and ϕ (in Fig. 1) as a coordinate for this task. To explain this in detail, suppose that the crank lies in the horizontal $(X-Y)$ plane (Fig. 1). If the hand grasps the handle of crank tightly, constraints r = const and Z = const are added to the system. Here, the variable Z indicates the vertical position of the hand. Therefore, to make the manipulator turn the crank, we must feedback only for ϕ azimuth. If we use the control law of equation (38), $\mathbf{y} - \bar{\mathbf{y}} = \phi - \bar{\phi}$, W is scalar and $J^T = (\partial\phi/\partial\mathbf{x})^T$ is an n dimensional vector. Since the vector (ϕ, r) is related with (X, Y) as

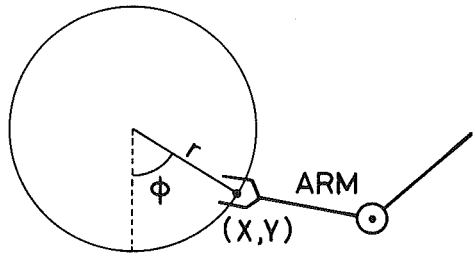


Fig. 1 Task-oriented coordinate for turning a crank

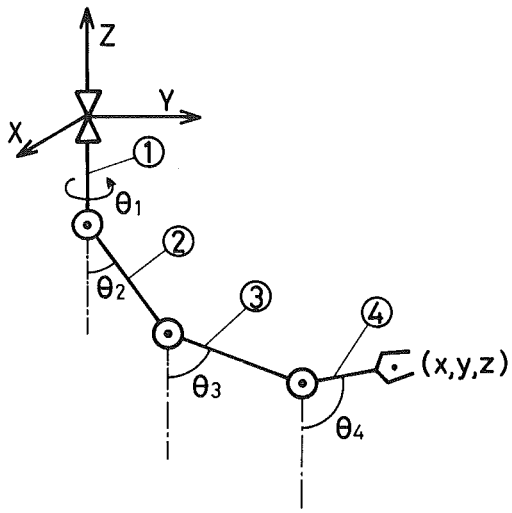


Fig. 2 Manipulator model with 4 degrees of freedom

$$\begin{pmatrix}\dot{X} \\ \dot{Y}\end{pmatrix} = \begin{pmatrix}r\cos\phi & \sin\phi \\ r\sin\phi & \cos\phi\end{pmatrix}\begin{pmatrix}\dot{\phi} \\ \dot{r}\end{pmatrix} = J_1\begin{pmatrix}\dot{\phi} \\ \dot{r}\end{pmatrix}$$

we can easily obtain the relation

$$\begin{pmatrix}\frac{\partial\phi}{\partial\mathbf{x}} \\ \frac{\partial r}{\partial\mathbf{x}}\end{pmatrix} = J_1^{-1}\begin{pmatrix}\frac{\partial X}{\partial\mathbf{x}} \\ \frac{\partial Y}{\partial\mathbf{x}}\end{pmatrix} = J_1^{-1}J_2 = \frac{1}{r}\begin{pmatrix}\cos\phi & -\sin\phi \\ r\sin\phi & r\cos\theta\end{pmatrix}J_2$$

and the Jacobian matrix $J = (\partial\phi/\partial\mathbf{x})$ as follows,

$$J = \left(\frac{\partial\phi}{\partial\mathbf{x}}\right) = \left(\frac{1}{r}\cos\phi, -\frac{1}{r}\sin\phi\right)J_2 \tag{51}$$

Incrementing the value of $\bar{\phi}$ by degrees, the manipulator will turn the crank by degrees.

It should be noted that the information for the position of the center of rotation is not necessary to turn the crank by this control law. Moreover, the control law can be used when the value of r is unknown, since the value of r can be put into the term of feedback gain matrix W.

5 Simulation Results

Effectiveness of the proposed control method was examined by computer simulation utilizing an arm model of Fig. 2. This arm has 4 degrees of freedom. Parameters of this arm are set in Table 1. We derived the equation of motion for this arm by Lagrange's method as follows,

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Table 1 Parameters of manipulator

Link	Mass (Kg)	I_x (Kgm^2)	I_y (Kgm^2)	I_z (Kgm^2)	Length (m)
1	4.0			0.04	
2	2.0	0.06	0.06	0.008	0.6
3	1.5	0.03	0.03	0.0025	0.5
4	1.0	0.01	0.01	0.001	0.3

$$\dot{\mathbf{x}} = R^{-1}(\mathbf{x})\mathbf{p} \tag{52}$$

$$\dot{p}_i = \frac{1}{2}\mathbf{p}^T R^{-1} R^i R^{-1}\mathbf{p} - g_i \sin\theta_i + u_i \quad (i=1, \ldots, 4) \tag{53}$$

where

$$\mathbf{x} = (\theta_1, \theta_2, \theta_3, \theta_4)^T, \ \mathbf{p} = (p_1, p_2, p_3, p_4)^T$$

The R is a 4×4 symmetric and positive definite matrix whose elements r_{ij} $(i,j = 1, \ldots, 4)$ are written as

$$\begin{aligned}
r_{11} &= (m_2 s_3^2 + m_3 l_2^2 + m_4 l_2^2 + I_{2y})(\sin\theta_2)^2 \\
&+ (m_3 s_3^2 + m_3 l_3^2 + I_{3y})(\sin\theta_3)^2 \\
&+ (m_4 l_4^2 + I_{4y})(\sin\theta_4)^2 \\
&+ 2(m_3 l_2 s_3 + m_4 l_2 l_3)\sin\theta_2 \cdot \sin\theta_3 \\
&+ 2m_4 l_3 l_4 \sin\theta_3 \cdot \sin\theta_4 \\
&+ 2m_4 l_2 s_4 \sin\theta_2 \cdot \sin\theta_4 \\
&+ I_{2z}(\cos\theta_2)^2 + I_{3z}(\cos\theta_3)^2 \\
&+ I_{4z}(\cos\theta_4)^2 + I_{1z} \\
r_{23} &= (m_3 l_2 s_3 + m_4 l_2 l_3)\cos(\theta_2 - \theta_3) + I_{3x} + I_{4x} \\
r_{24} &= m_4 l_2 s_4 \cos(\theta_2 - \theta_4) + I_{4x}, \\
r_{34} &= m_4 l_3 s_4 \cos(\theta_3 - \theta_4) + I_{4x} \\
r_{12} &= r_{13} = r_{14} = 0, \\
r_{22} &= m_2 s_2^2 + (m_3 + m_4) l_2^2 + I_{2x} + I_{3x} + I_{4x} \\
r_{33} &= m_3 l_3^2 + m_4 l_3^2 + I_{3x} + I_{4x}, r_{44} \\
&= m_4 s_4^2 + I_{4x}, r_{ij} = r_{ji}
\end{aligned}$$

where m_i is the mass of the ith link and l_i is the length from the ith joint to the $(i+1)$th joint, and s_i is the length from the ith joint to the center of mass of the ith link. The s_i is set as $s_i = l_i/2$. I_{iA} is the moment of inertia for A axis of the ith link $(A = x, y, z)$. The matrix R^i is defined as $R^i = (\partial R/\partial\theta_i)$ $(i = 1, \ldots, 4)$. Coefficients g_i $(i = 1, \ldots, 4)$ are $g_1 = 0$, $g_2 = (m_2 s_2 + m_3 l_2 + m_4 l_2)\, g$, $g_3 = (m_3 s_3 + m_4 l_2)\, g$, $g_4 = m_4 s_4 g$ where g is a gravitational acceleration and is set $g = 9.8$ m/s^2. The generalized forces u_i $(i = 1, \ldots, 4)$ are related with the joint torques τ_i $(i = 1, \ldots, 4)$ as follows,

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \\ \tau_4 \end{pmatrix}$$

This nonlinear differential equation was used for the simulation.

Only the results for the case of task-oriented coordinate control with control law of equation (32) are shown in the following. First, we chose spatial coordinates (x, y, z) of a tip of the 4th link and θ_4 as task-oriented coordinates, that is, $\mathbf{y}$ $= (x, y, z, \theta_4)$. In Fig. 3 and Fig. 4 an initial position of joint angles is set as

$$(\theta_1, \theta_2, \theta_3, \theta_4) = (0, -0.3, 1.2, 1.57)\ (\text{rad})$$

A target position $\bar{\mathbf{y}}$ is given as $\bar{\mathbf{y}} = \mathbf{y}(0) + \Delta\mathbf{y}$. In Fig. 3 $\Delta\mathbf{y}$ is set as $\Delta\mathbf{y}^T = (0.4\text{m}, 0.3\text{m}, 0.2\text{m}, 0\ \text{rad})$. Feedback gain matrices W and Q of equation (32) are set as

$$W = \text{diag}\,\{30, 30, 30, 5\},\ Q = \text{diag}\,\{15, 20, 15, 5\}$$

These feedback gains were chosen after several trials. Vector $\mathbf{y}$ converges to the target smoothly in about 2 seconds. In this case, control $\mathbf{u}$ is large at the beginning. If we set a trajectory which joins the initial position and the final position and change the intermidiate target position $\bar{\mathbf{y}}$ along this trajectory, then the initial large magnitude of $\mathbf{u}$ can be reduced. In Fig. 4

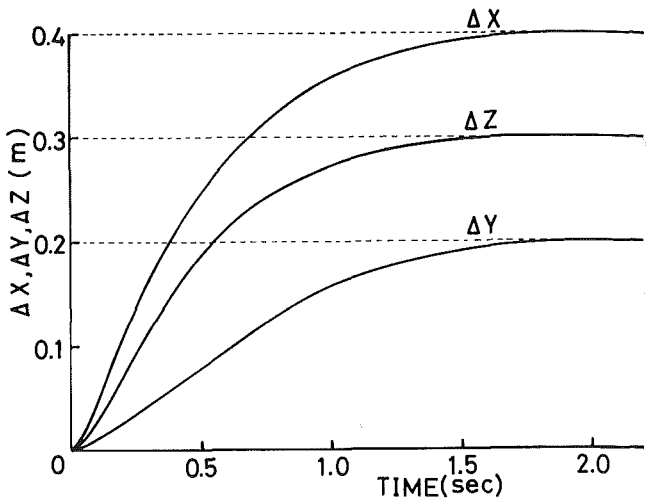


Fig. 3 System response with setting a target position

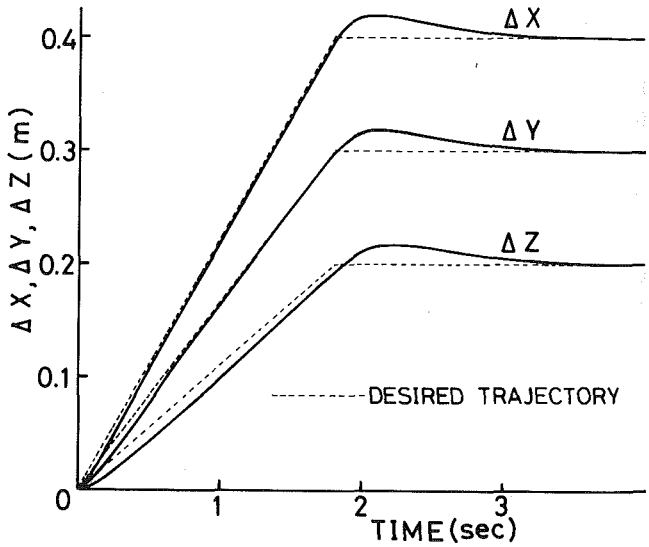


Fig. 4 System response with setting a desired trajectory

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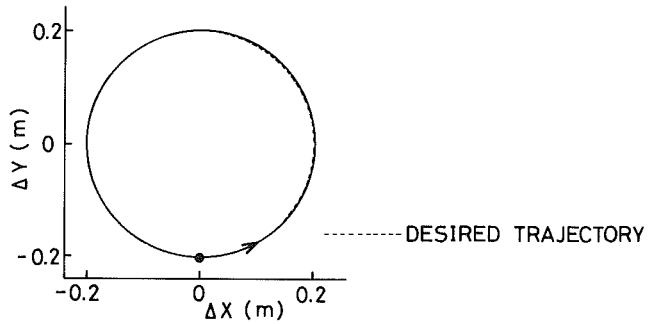


Fig. 5 Response of a task-oriented coordinate for turning a crank (1) trace of Hand's position in X-Y Plane

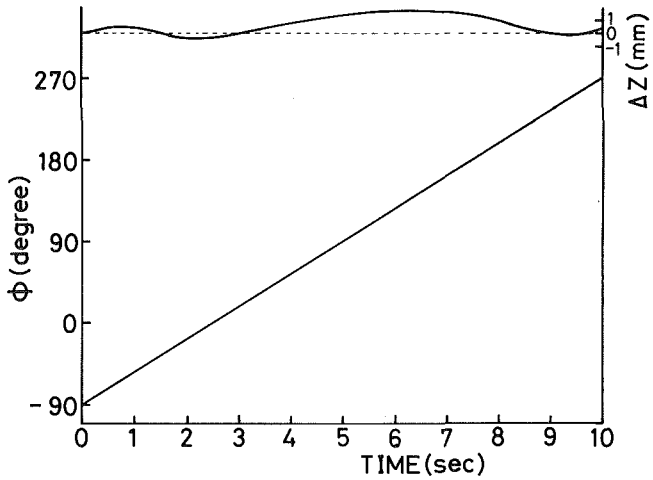


Fig. 6 Response of a task-oriented coordinate for turning a crank (2) time response of ϕ and deviation for Z axis

we set a desired trajectory $\bar{\mathbf{y}}(t)$ (a dotted line). In this case the control law is set as

$$\mathbf{u} = \left(\frac{\partial V}{\partial \mathbf{x}}\right)^T - J^T\{W(\mathbf{y} - \bar{\mathbf{y}}(t)) + Q\dot{\mathbf{y}}\}$$

and same feedback gain matrices are used. Because the velocity of the desired trajectory is not continuous, overshoot appears as shown in the figure. This overshoot can be suppressed by modifying the desired trajectory.

Next, in order to simulate the task of turning a crank, we chose $\mathbf{y} = (r, \phi, z, \theta_4)^T$ where r and ϕ are defined in Fig. 1. A desired trajectory is determined so that ϕ makes one revolution per 10 seconds while r = const and z = const are maintained. Initial joint angles are set as

$$(\theta_1, \theta_2, \theta_3, \theta_4) = (0, -0.7, 1.5, 1.57)\ (\text{rad})$$

Gain matrices W and Q are set as

$$W = \text{diag}\{200, 5, 15, 3\},\ Q = \text{diag}\{200, 5, 15, 3\}$$

where feedback gains for r axis are chosen to be quite large in comparison with other axes since the system is subject to constraint in the direction of r axis. As shown in Fig. 5 and Fig. 6, the arm traces the desired trajectory almost completely. On the other hand, we attempted a feedback control by use of only a part of task-oriented coordinate $\mathbf{y}$. However, the response by this method is inferior to that by a feedback of all coordinates.

Though the asymptotic stability of the system is assured by fixed feedback gains as shown in sections 2 and 3, several trials for selecting gains are necessary in order to obtain better response. However, so far as these simulation results show, it seems that a single set of gains which shows a good response to a given standard task can be used successfully in a large region of work space.

6 Conclusion and Discussion

A theoretical consideration of the motion control of manipulators has been presented. Simple and useful control methods for nonlinear dynamics of general mechanical systems have been proposed from the view point of mechanics. This approach has shown its power in the task-oriented coordinate control and in the case of constrained dynamics. The proposed task-oriented coordinate control method seems suitable to sensor feedback control because the real data from each sensor can be directly used as a feedback input.

The change of environment around the manipulator may often produce mechanical constraints which may be described with much uncertainty. In many cases, the singularity of the Jacobian matrix is not avoidable in a wide range of motion and influenced by the constraints. In these circumstances, a feedback control of only part of coordinates which are attainable under the constraints seem to be an efficient method. In fact, our approach verifies the validity of this method.

In this paper, only the most basic studies on the motion control of general manipulators are carried out and improvement of response of the motion have not been sufficiently discussed. To obtain a highly refined motion, we should tune the feedback gains taking into account the matrix $R(\mathbf{x})$. Neverthless, the simulation results show that a desirable motion can be obtained by choosing the gain matrices W and Q appropriately. More advanced control such as tracking a trajectory whose acceleration can not be neglected are under study for the system of task-oriented coordinate.

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